



CS & IT ENGINEERING



Algorithms

Analysis of Algorithms

Lecture No.- 06



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

→ Asymptotic Notations
→ Properties

Topics to be Covered



Topic

Topic

Topic

Properties

Examples



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



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TTTe



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(1) If $0 < x < y$, then $n^x = O(n^y)$

True

$$\begin{array}{l} x < y \\ n^x < n^y \Rightarrow n^2 = O(n^3) \\ \vdots \\ n^3 = O(n^4) \end{array}$$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(2) $\log(n)$ is $\Omega(1/n)$

True

$$\log n > \frac{1}{n}$$

$$\log n = \Omega(1/n) \checkmark$$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(3) 2^{n^2} is $O(n!)$

False

$$\begin{array}{ccc} 2^{n^2} & > & n^n \\ \log_2(2^{n^2}) & & \log(n^n) \\ n^2 & & n \log n \end{array}$$

$$n \times n \quad \cancel{n \log n}$$

$$n > \log n$$



Topic : General Properties of Big Oh Notation



Practice Question (T/F):

$$(4) 20 \cdot n \cdot \log n = O(n \log n)$$



True



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(5) $(n+c)^k \neq O(n^k)$ for some $k > 0, c > 0$

False

$$\begin{array}{l} (n+c)^k \\ \hline (n+2)^k \\ (n+5)^3 \end{array} \longrightarrow \text{poly of degree } \underline{k}.$$

$O(n^k)$



Topic : General Properties of Big Oh Notation

Practice Question (T/F):

(6) n^2 is $O(2^{\log n})$

$$n^2 > 2^{\log n}$$

False

$\Rightarrow$

maths $\rightarrow$

$$\begin{array}{cc} n^2 & 2^{\log n} \\ \hline \log(n^2) & \log(2^{\log n}) \\ 2 \log n & > \log n \end{array}$$

$$\begin{array}{l} 2^{n^2} = 4^n ? \\ 2^9 = 4^3 \end{array}$$

Shortcut: $n^2 \cdot 2^{(\log n)} = 2^{\log_2 n}$

$$n^2 > n^{\log_2 2} = n'$$



Topic : Adding Functions



The sum of two functions is governed by the dominant one, namely:

$$\underline{O(f(n))} + O(g(n)) \rightarrow O(\max(f(n), g(n)))$$

$$\underline{\Omega f(n)} + \Omega(g(n)) \rightarrow \Omega(\max(f(n), g(n)))$$

$$\underline{\theta(f(n))} + \theta(g(n)) \rightarrow \theta(\max(f(n), g(n)))$$



Topic : Adding Functions

Example:

$$f(n) = 5n^2 + 2 \rightarrow O(n^2)$$

$$g(n) = 10n^3 \rightarrow O(n^3)$$

$$O(f(n)) = n^2 \text{ and } O(g(n)) = n^3$$

$$n^2 + n^3 = O(\max(5n^2 + 2, 10n^3))$$

$$= O(10n^3)$$

$$\neq O(n^3) \rightarrow \Omega(n^3) \rightarrow \underline{O(n^3)}$$



Topic : Adding Functions



$$O(f(n)) * O(g(n)) \rightarrow O(f(n) * g(n))$$

$$\Omega(f(n)) * \Omega(g(n)) \rightarrow \Omega(f(n) * g(n))$$

$$\theta(f(n)) * \theta(g(n)) \rightarrow \theta(f(n) * g(n))$$





Topic : General Properties of Big Oh Notation

Imp. Practice Question (T/F):

(1) $n^2 = O(2^{(2 \log n)})$

True

V. Imp

$$\begin{array}{r} n^2 \quad \bar{A} \quad 2^{2 \log n} \\ \hline \log(n^2) \quad \log(2^{2 \log n}) \end{array}$$

$$2 \log n = 2 \log n$$

$$\underline{\underline{n^2 = \Theta(2^{2 \log n})}}$$

Shortcut

$$n^2$$

$$2^{2 \log_2 n}$$

$$= n^{2 \log_2 2}$$

$$n^2$$

$$= n^2$$



Topic : General Properties of Big Oh Notation

Imp. Practice Question (T/F):

$$(2) (\log n)^{1/2} = O(\log(\log n))$$

False

$$(\log n)^{1/2}$$

$$\frac{1}{2} \log(\log n)$$

>

$$\log(\log n)$$

$$\log(\log(\log n))$$

$$(\log n)^{1/2} > \log(\log n)$$

$$\text{Let } \underline{\log n = n}$$

$$n^{1/2} > \log(n)$$



Topic : General Properties of Big Oh Notation

Imp. Practice Question (T/F):

(3) $a^n \neq O(n^x)$, for $a > 1, x > 0$

True

$\text{Expo} \neq O(\text{poly})$

$a^n > n^x$
 $\downarrow \quad \downarrow$
 $\text{Expo} > \text{poly}$



Topic : Trichotomy Property

#Q. Does Asymptotic notations follow Trichotomy Property?

→ No



Topic : Trichotomy Property

Trichotomy Property in Real Numbers:

- Given any two real numbers x & y (fixed).

Then, x and y follow exactly one of the below relation:

1) $x = y$ or,

2) $x > y$ or,

3) $x < y$

5 7
 $5 < 7$



Topic : Trichotomy Property

Trichotomy Property in Asymptotic Comparison of two function:

- Given two functions $f(n)$ & $g(n)$, does $f(n)$ & $g(n)$ always follow exactly one of the below?

$$1) f(n) >_A g(n) \quad \Rightarrow \quad \underline{F = \Omega(g), \omega(g)}$$

$$\text{or, } 2) f(n) =_A g(n) \quad \Rightarrow \quad \underline{F = \theta(g)}$$

$$\text{or, } 3) f(n) <_A g(n) \quad \Rightarrow \quad \underline{F = O(g), o(g)}$$



Topic : Trichotomy Property



Example1:

$$f(n) = 5n^2$$

$$g(n) = 7n^3$$

$$5n^2 < 7n^3$$

$$\underline{f = O(g)}$$



Topic : Trichotomy Property



Example2:

$$f(n) = 5n^2$$

$$g(n) = 1/n$$

$$5n^2 \quad \frac{1}{n}$$

$$f > g$$

$$\boxed{f = \Omega(g)}$$



Topic : Trichotomy Property



Example3:

$$f(n) = 10n^3 + 7$$

$$g(n) = 5n^3 + 15$$

$$f = 10n^3 + 7 = \Theta(n^3)$$

$$g = 5n^3 + 15 = \Theta(n^3)$$

$$\boxed{f = g}$$



Topic : Trichotomy Property



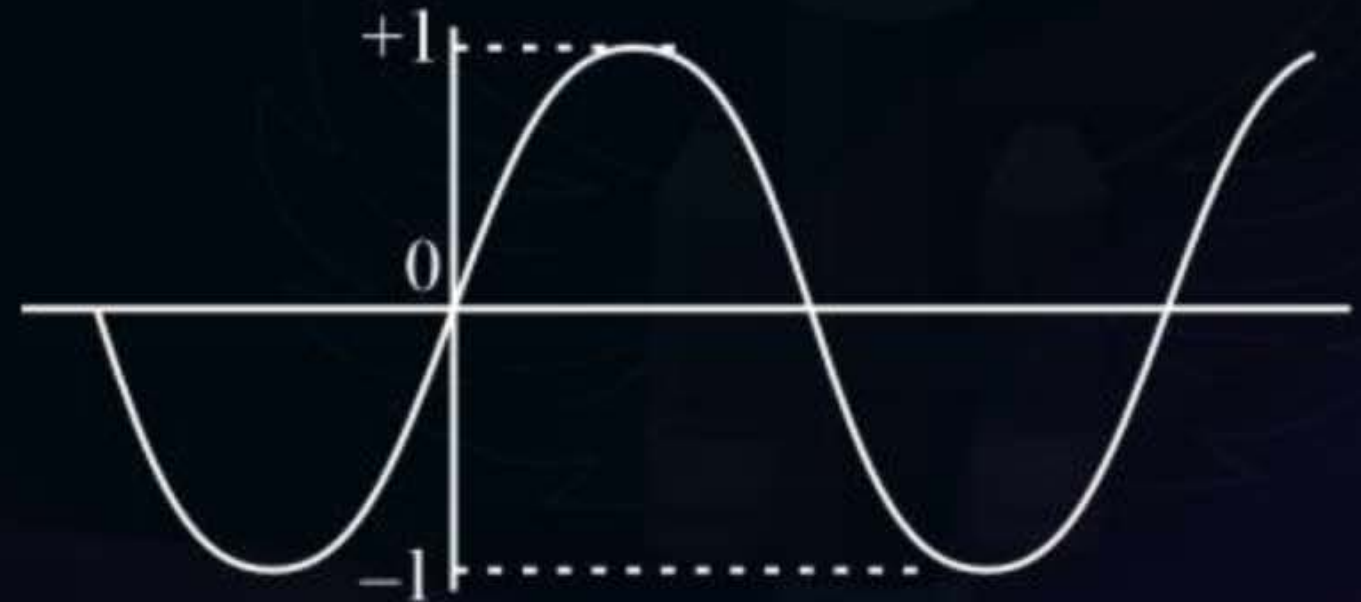
Example4:

P4Q

$$f(n) = n$$

$$g(n) = n^{(1+\sin x)}$$

Sin x plot (wave)



$$f(n) = n$$

$$g(n) = n^{(1 + \sin x)}$$

Case 1: $\sin x = 1$

$$f(n) = n$$

$$g(n) = n^{(1+1)} = n^2$$

$$\boxed{f = O(g)} \checkmark$$

$$\underline{n = O(n^2)}$$

Case 2: $\sin x = -1$

$$f(n) = n$$

$$g(n) = n^{(1+(-1))} = n^0 = 1$$

$$f > g \Rightarrow \boxed{f = \Omega(g)} \checkmark$$



Topic : Trichotomy Property

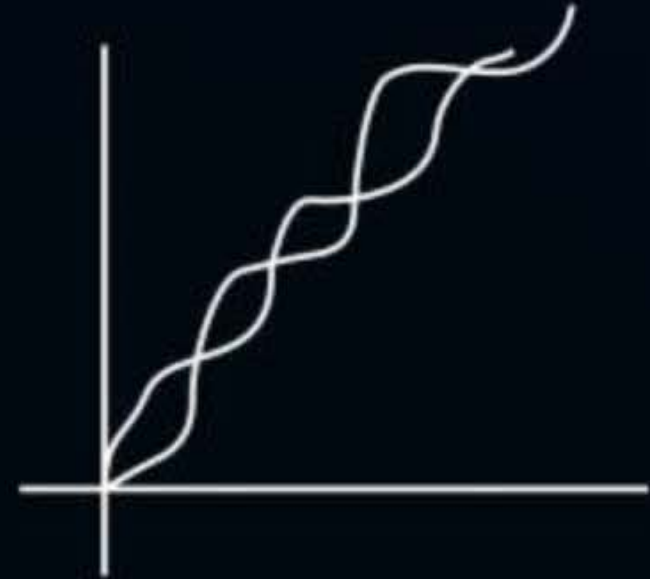


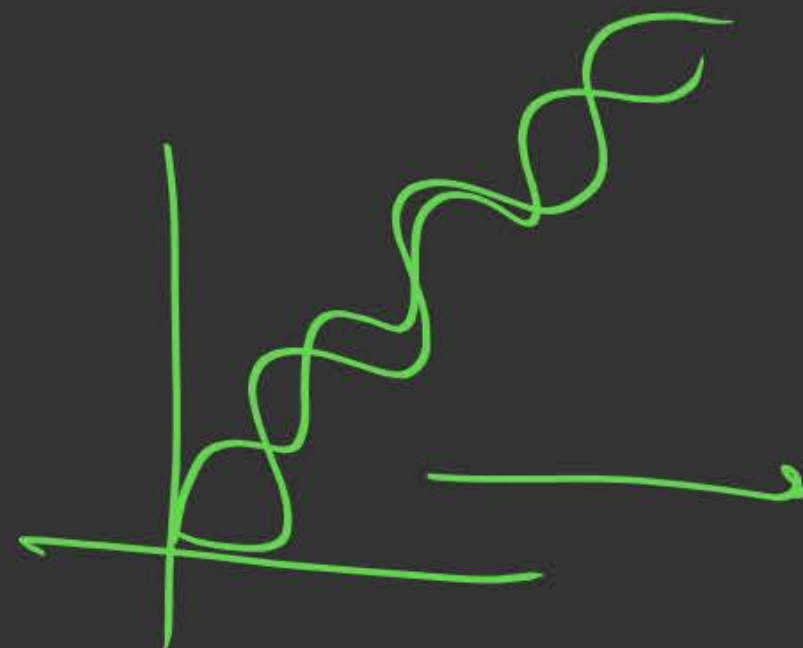
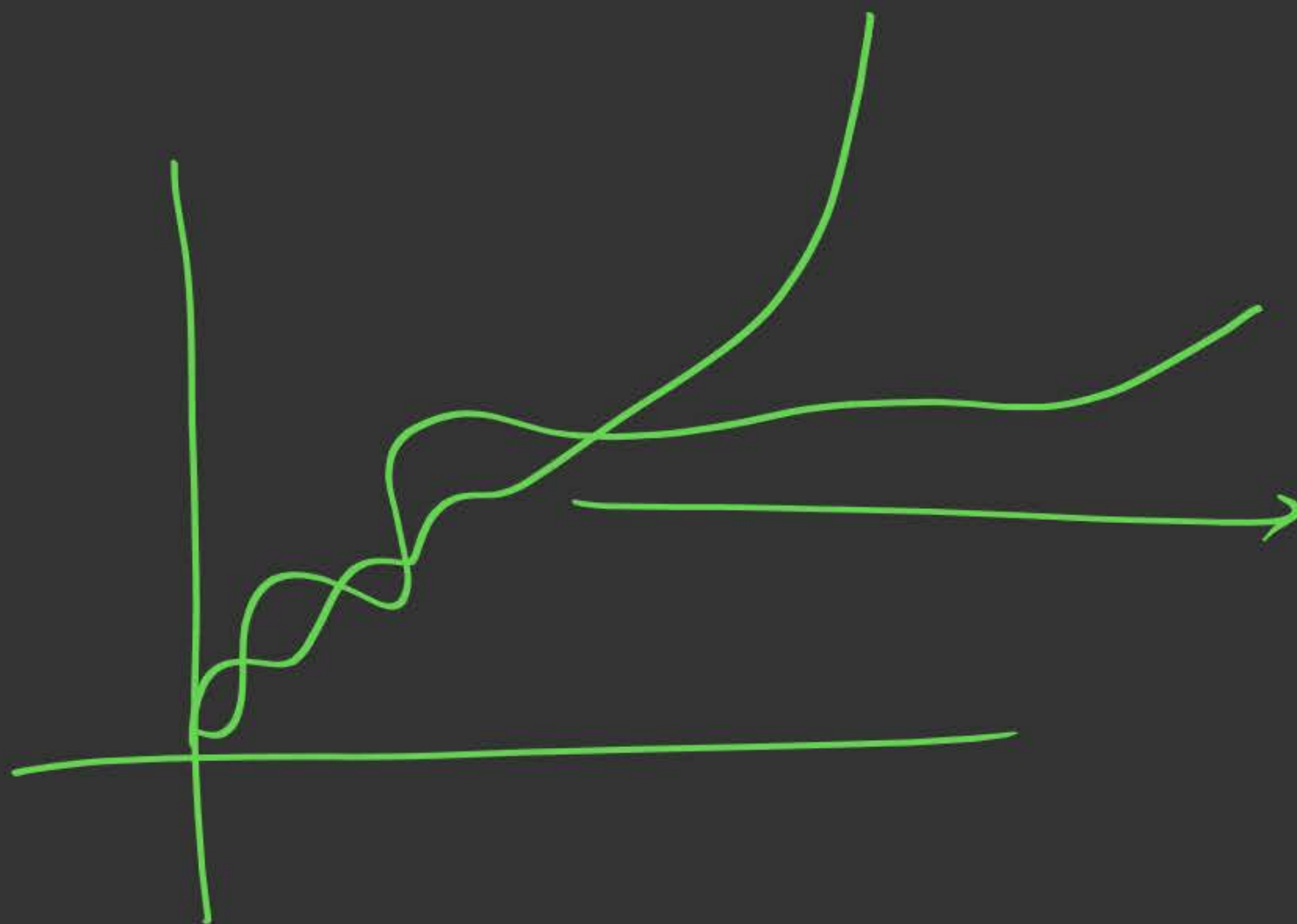
Observation from previous equation:

$$f(n) = n$$

$$g(n) = n^{(1+\sin x)}$$

- $f(n)$ & $g(n)$ are not holding the trichotomy property as they can't get a clear asymptotic comparison between both for $n \geq n_0$.







Topic : Trichotomy Property



Conclusion:

- Asymptotic notations may or may not follow Trichotomy Property (holds sometimes)
- Hence, property does not hold

(As property holds if its always satisfied for all cases)

$$(a+b)^2 = \underline{a^2 + 2ab + b^2}$$



Topic : Analysis of Algorithms

Discrete Properties of Asymptotic Notations:

$\equiv$

	Reflexive	Symmetric	Transitive	Transpose Symmetry
O	✓	x	✓	✓
Ω	✓	x	✓	✓
θ	✓	✓	✓	x
o	x	x	✓	✓
ω	x	x	✓	✓

$$a < b < c \Rightarrow a < c$$

$$F = O(g)$$

$$g = \Omega(F)$$



Topic : Analysis of Algorithms



Transitive:

$$a \leq b \ \& \ b \leq c \Rightarrow a \leq c$$



Topic : Analysis of Algorithms

Let a, b be two real numbers and f, g be two functions of n .

(1) If $f(n) = O(g(n)) \Rightarrow a \leq b$

(2) If $f(n) = \Omega(g(n)) \Rightarrow a \geq b$

(3) If $f(n) = \theta(g(n)) \Rightarrow a = b$

(4) If $f(n) = o(g(n)) \Rightarrow a < b$

(5) If $f(n) = \omega(g(n)) \Rightarrow a > b$

Adv Questions

[MCQ]

#Q. $f(n), g(n)$: are functions

given: $f(n) = O(g(n))$

(a) Is $f(n) = O(f(n)^2)$ $\longrightarrow$ **F**

(b) $2^{f(n)} = O(2^{g(n)})$ $\longrightarrow$ **F**

A

Only (a) is true

B

Only (b) is true

C

Both (a) & (b) are true $\longrightarrow 74\%$

D

Both (a) & (b) are false $\longrightarrow 0\%$

$$\text{Is } f(n) = O(f(n)^2) ?$$

$$\text{eg-1)} \quad f(n) = n$$

$$f(n)^2 = n^2$$

$$n = O(n^2)$$

$$\underline{f(n) = O(f(n)^2)}$$

$$\text{eg2)} \therefore f(n) = 1/n$$

$$f(n)^2 = \left(\frac{1}{n}\right)^2 = \frac{1}{n^2}$$

$$\frac{1}{n} > \frac{1}{n^2}$$

$$f(n) = O(f(n)^2) ? \quad \text{No}$$

(B)

$$\text{given } f(n) = O(g(n))$$

$$\text{Is } 2^{f(n)} = O(2^{g(n)})$$

$$\text{eg 1) let } \left. \begin{array}{l} f(n) = n \\ g(n) = n^2 \end{array} \right\} \checkmark$$

$$2^n = O(2^{n^2})?$$

Yes

$$\text{eg 2: } \left. \begin{array}{l} f(n) = n \\ g(n) = 10 \end{array} \right\} ?$$

invalid

$$\underline{f(n) = O(g(n))} \times$$

Q3) $f(n) = 2n$
 $g(n) = n$

$f(n) = O(g(n))$? ✓

$2^{f(n)} = O(2^{g(n)})$?

$2^{(2n)} = O(2^n)$

$4^n = O(2^n)$?

No

[MCQ]

#Q. Which one of the following statements is True for all positive functions $f(n)$?

46.6%

A

A

$f(n^2) = \theta(f(n)^2)$, when $f(n)$ is a polynomial

B

$f(n)^2 = O(f(n)^2)$ *small oh* ~~X~~

C

$f(n)^2 = O(f(n)^2)$, when $f(n)$ is an exponential function ~~X~~

D

$f(n^2) = \Omega(f(n)^2)$ ~~X~~

A)

$$f(n) = n^2$$

$$\bar{f}(n)^2 = (n^2)^2 = n^4$$

$$f(n^2) = (n^2)^2 = n^4$$

$$\underline{f(n^2) = O(\bar{f}(n)^2)}$$

③

$$f(n) = n^2$$

$$f(n)^2 = n^4$$

$$f(n^2) = n^4$$

<

$$n^4 = O(n^4) ?$$

No

©

$$f(n) = 2^n$$

$$F(n^2) = 2^{(n^2)}$$

$$f(n)^2 = (2^n)^2 = 2^{2n}$$

$$2^{n^2} > 2^{2n}$$

①

$$f(n) = \log n$$

$$f(n^2) = \log(n^2) = 2 \log n$$

$$(f(n))^2 = (\log n)^2$$

$$2 \log n \geq (\log n)^2$$

No



Summary



→ Properties
→ Practice



THANK - YOU